

Multi-Neighbour Model Selection Report

June 19, 2026

1 Introduction

This report documents the simulation-based selection of synchronisation models for a decentralised multi-neighbour drone network. The purpose of this stage is not to claim a globally optimal synchronisation algorithm, but to select suitable candidate models for the next hardware-in-the-loop (HIL) stage under the tested simulation conditions.

Five candidate models were compared: a continuous Kuramoto phase-coupling baseline, two pulse-coupled oscillator (PCO) variants, and two event-based adaptive phase/frequency locking (EAPF) variants. The evaluation first locks model parameters using a validation sweep, then compares the locked models using independent random seeds, N=3 HIL-aligned tests, N=5 scalability tests, and a weighted engineering selection matrix.

2 Candidate Models

This section defines the five models used in the final evaluation. All models produce visible flash outputs, and all are evaluated using the same flash-based synchronisation metrics. However, their internal coupling mechanisms differ. The PCO and EAPF variants are event-driven: they update their internal state in response to detected neighbour flash events. The Kuramoto baseline uses continuous phase coupling in simulation and would require phase reconstruction from flash timestamps before hardware implementation.

Common terminology. Each agent has an internal oscillator state and emits a flash when its oscillator reaches the firing condition. A *flash event* is the observable timing output available to other agents. A *neighbour* is another agent visible through the selected topology graph. *Zero-lag synchronisation* means that agents flash nearly simultaneously. *Phase-locked synchronisation* permits a stable non-zero timing offset. *One-to-one flash locking* means that each agent produces one flash per group cycle, avoiding extra flashes or harmonic locking.

2.1 Kuramoto: continuous phase coupling

Core idea. The Kuramoto model represents each agent as a continuously rotating phase oscillator. Each oscillator adjusts its instantaneous phase velocity according to the phase difference between itself and its visible neighbours. If a neighbour is ahead in phase, the coupling term tends to accelerate the local oscillator; if the local oscillator is ahead, the coupling term tends to slow it down. This bidirectional acceleration/deceleration property makes Kuramoto a strong baseline for heterogeneous oscillator networks [1].

Update rule. For agent i , the phase dynamics are

$$\dot{\theta}_i = \omega_i + \frac{K}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} \sin(\theta_j - \theta_i), \quad (1)$$

where θ_i is the phase of agent i , ω_i is its natural angular frequency, $\mathcal{N}_i$ is the set of visible neighbours, and K is the coupling gain.

Locked parameter. The parameter-locking sweep selected $K = 5.0$.

Expected strengths and limitations. Kuramoto is mathematically transparent, compact, and naturally supports multiple neighbours through a summation over phase differences. Its main limitation for this project is that the simulation assumes continuous phase information, whereas the planned visual hardware observes only flash timestamps. Therefore, hardware use would require flash-to-phase reconstruction or phase estimation.

2.2 PCO simple: event-driven pulse coupling

Core idea. The simple PCO model is the most direct firefly-inspired baseline. Each agent has a normalised phase $\phi_i \in [0, 1)$. The phase increases at the natural frequency. When ϕ_i reaches 1, the agent flashes and resets. When it detects a neighbour flash, it advances its own phase by a fixed amount. This captures the idea that observed flashes can pull an oscillator toward earlier firing, inspired by pulse-coupled integrate-and-fire oscillator models [2, 3] and firefly-inspired synchronisation in ad hoc networks [4].

Update rule. The locked simple PCO variant uses additive phase advance:

$$\phi_i \leftarrow \min(1, \phi_i + \varepsilon). \quad (2)$$

If the updated phase reaches the threshold, the agent flashes, resets, and enters a refractory period.

Locked parameters. The locked parameters are additive phase coupling, $\varepsilon = 0.10$, and refractory period $\tau_{\text{ref}} = 0.05$ s.

Expected strengths and limitations. The model is simple, event-driven, and conceptually close to visual firefly flashing. However, because the locked simple variant mainly advances phase, it cannot reliably slow down an oscillator whose natural frequency is too high. This can cause extra flashes under frequency heterogeneity.

2.3 PCO adaptive PRC: phase response curve variant

Core idea. The adaptive PRC variant extends the simple PCO by replacing fixed phase advance with a phase response curve (PRC). A PRC defines how a received pulse affects the receiver depending on its current phase. Unlike the simple phase-advance rule, a biphasic PRC can either advance or delay the oscillator. This was added to reduce extra flashes from fast agents in heterogeneous networks.

Update rule. The locked adaptive PRC variant uses a biphasic sine correction:

$$\Delta\phi_i = \varepsilon \sin(2\pi(\phi_i - \phi_{\text{ref}})), \quad (3)$$

with negative corrections allowed. Positive $\Delta\phi_i$ advances the next flash, while negative $\Delta\phi_i$ delays it. The correction is limited to a maximum magnitude.

Locked parameters. The locked parameters are biphasic sine PRC, $\varepsilon = 0.10$, phase delay enabled, and maximum phase correction $\Delta\phi_{\text{max}} = 0.20$.

Expected strengths and limitations. The adaptive PRC is more flexible than the simple PCO because it can partly counteract fast oscillators. However, it is less biologically simple and introduces additional tuning parameters. It is retained as an engineering extension of the firefly-inspired model rather than as an exact reproduction of any single biological model.

2.4 EAPF tracker: event-based phase/frequency tracking

Core idea. The EAPF tracker was originally developed for fixed-leader entrainment. A detected flash is treated as a timing reference. The agent computes its own phase error at the flash time and applies phase and frequency corrections. In a multi-neighbour setting, this tracker can be applied to neighbour flash events, but it does not explicitly maintain a distributed consensus state. It therefore provides a useful comparison between leader-tracking control and mutual multi-agent synchronisation.

Update rule. For a detected reference flash, the tracker computes

$$e_\theta = \text{wrap}_\pi(0 - \theta_i), \quad (4)$$

then updates

$$\theta_i \leftarrow \text{wrap}_{2\pi}(\theta_i + g_p e_\theta), \quad (5)$$

$$f_i \leftarrow \text{clamp}\left(f_i + g_f \frac{e_\theta}{2\pi}, f_{\min}, f_{\max}\right). \quad (6)$$

This is related to phase-locking and PLL-style correction methods [6].

Locked parameters. The locked parameters are phase gain $g_p = 0.40$ and frequency gain $g_f = 0.15$.

Expected strengths and limitations. The tracker is event-based and effective for leader-following or near-identical networks. Its limitation is that it treats detected flashes as references without a full multi-neighbour consensus mechanism, making it less robust under strong heterogeneity and sparse directed topologies.

2.5 EAPF consensus: event-based phase/frequency consensus

Core idea. The EAPF consensus model extends event-based phase/frequency locking into a distributed multi-neighbour agreement mechanism. Each agent records neighbour flash timestamps, estimates neighbour periods and frequencies from those timestamps, propagates estimated neighbour phases between flashes, and then slowly adjusts its own phase and frequency toward the local neighbourhood average. This is an event-based engineering analogue of distributed consensus in multi-agent systems [5], combined with phase/frequency locking ideas [6].

Update rule. For agent i , the consensus errors are

$$e_{\theta,i} = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} \text{wrap}_\pi(\hat{\theta}_j - \theta_i), \quad (7)$$

$$e_{f,i} = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} (\hat{f}_j - f_i), \quad (8)$$

where $\hat{\theta}_j$ and $\hat{f}_j$ are locally estimated neighbour phase and frequency values derived from flash timestamps. The phase and frequency updates are low-pass filtered and rate-limited to avoid overshoot and correction-induced extra flashes.

Locked parameters. The locked parameters are $g_p = 0.02$, $g_f = 0.02$, phase/frequency filter coefficients $\alpha_p = \alpha_f = 0.2$, maximum phase step $\Delta\theta_{\max} = 0.2$ rad, and maximum frequency step $\Delta f_{\max} = 0.05$ Hz.

Expected strengths and limitations. The model is well aligned with flash-event sensing because it does not require direct access to true neighbour phase. It can correct both slow and fast oscillators by adjusting phase and frequency. Its main limitations are increased algorithmic complexity, per-neighbour state estimation, and the need to validate robustness under real camera latency and missed detections.

Table 1: Locked model parameters selected via a $60 \text{ s} \times 5$ -repeat parameter-locking sweep using validation seeds 1000–1004.

Model	Locked parameters	Variant
Kuramoto	$K = 5.0$	baseline
PCO simple	additive phase, $\varepsilon = 0.10$, $\tau_{\text{ref}} = 0.05 \text{ s}$	simple
PCO adaptive PRC	biphasic sine PRC, $\varepsilon = 0.10$, phase delay on, $\Delta\phi_{\max} = 0.20$	adaptive PRC
EAPF tracker	$g_p = 0.40$, $g_f = 0.15$	tracker
EAPF consensus	$g_p = 0.02$, $g_f = 0.02$, $\alpha_p = \alpha_f = 0.2$, $\Delta\theta_{\max} = 0.2 \text{ rad}$, $\Delta f_{\max} = 0.05 \text{ Hz}$	consensus

3 Evaluation

3.1 Simulation setup

This evaluation uses a discrete-time multi-agent simulator to compare the synchronisation models under controlled and repeatable conditions. Each simulated drone is represented as an oscillator-based agent with an internal timing state, a natural flashing frequency, and a visible flash output. The simulator advances all agents using a fixed time-step of $dt = 0.01$ s, corresponding to 6000 simulation steps in each 60 s trial.

Neighbour visibility is represented by a graph-based adjacency structure rather than by a geometric camera field of view. The adjacency list maps each agent to the agents it can observe. In other words, an entry $i \rightarrow j$ means that agent i can detect the flash events produced by agent j . This convention is important because the observation edge and the influence direction are opposite: if agent i observes agent j , then j 's flashing can influence the oscillator update of i .

This graph abstraction deliberately removes camera image formation, field-of-view geometry, lighting variation, occlusion, and detection failure from the comparison. The purpose of this stage is therefore to compare the synchronisation algorithms under identical local-interaction graphs before reintroducing perception and hardware effects in the later HIL experiments.

Visibility topology definitions. The tested topologies define which neighbouring flashes each agent can observe. For the N=3 evaluation, three topologies are used:

- **all_to_all:** each agent observes the other two agents,

$$0 : [1, 2], \quad 1 : [0, 2], \quad 2 : [0, 1].$$

This represents the best-case mutual-visibility condition.

- **chain:** a bidirectional local chain,

$$0 : [1], \quad 1 : [0, 2], \quad 2 : [1].$$

The two endpoint agents observe only the middle agent, while the middle agent observes both endpoints. This tests propagation through a connected but non-fully connected local network.

- **directed_chain:** a one-way visibility chain,

$$0 : [], \quad 1 : [0], \quad 2 : [1].$$

Agent 0 observes nobody and is therefore effectively uninfluenced. Agent 1 observes agent 0, and agent 2 observes agent 1. Information therefore propagates in the direction $0 \rightarrow 1 \rightarrow 2$.

For the N=5 scalability evaluation, three sparse local-neighbour topologies are used:

- **local_ring_5:**

$$0 : [1, 4], \quad 1 : [0, 2], \quad 2 : [1, 3], \quad 3 : [2, 4], \quad 4 : [3, 0].$$

This represents a closed local-neighbour loop in which each agent observes two neighbours.

- **chain_5:**

$$0 : [1], \quad 1 : [0, 2], \quad 2 : [1, 3], \quad 3 : [2, 4], \quad 4 : [3].$$

This extends the propagation-chain test to five agents.

- **local_degree_2_3:**

$$0 : [1, 2], \quad 1 : [0, 2, 3], \quad 2 : [0, 1, 3], \quad 3 : [1, 2, 4], \quad 4 : [2, 3].$$

This represents a sparse local cluster in which each agent observes two or three nearby neighbours.

These topology definitions allow the same synchronisation model to be tested under best-case mutual visibility, local propagation, asymmetric information flow, and larger sparse-neighbour networks.

Per-step simulation order. At each time-step t , the simulator loops over all agents. For each agent, it first computes the model-specific neighbour input using the current topology. For event-driven models such as PCO and EAPF, this input is a set or count of visible neighbour flash events. For the Kuramoto baseline, this input is the continuous phase-coupling term computed from the true internal phases of visible neighbours in simulation:

$$\sum_{j \in \mathcal{N}_i} \sin(\theta_j - \theta_i).$$

The agent is then stepped forward by dt , and any resulting flash event is recorded. After all agents have been stepped, the simulator logs the phase, frequency, firing state, and flash events, then advances the global simulation time.

Event-based neighbour flashes are delivered within the same time-step according to the sequential update order. The firing state array is updated as agents are stepped, so if an earlier-indexed visible neighbour fires during the current step, that flash can be observed by later-indexed agents in the same step. This is a simulation implementation detail and should be interpreted as idealised zero-latency event delivery.

Model-specific information assumptions. Although all models are evaluated using the same topology graph and flash-output metrics, they do not use identical internal information. The Kuramoto baseline uses true neighbour phase values in the pure simulation, which is appropriate for testing the continuous phase-coupling model but would require flash-to-phase reconstruction in hardware. By contrast, the EAPF consensus model updates neighbour state only from detected flash timestamps. It records the time of neighbour flashes, estimates neighbour periods and frequencies from flash intervals, and propagates estimated neighbour phase locally between flashes. It therefore does not access another agent’s true internal phase or frequency.

Trial initialisation and randomisation. Each trial uses the natural frequencies defined by the selected frequency set. Initial oscillator phases are sampled independently from a uniform distribution over $[0, 2\pi)$ for each agent and each trial. The only source of randomness in the final simulation is this random initial phase assignment. No event noise, missed detections, or random delay is included. Each trial uses a deterministic random seed of the form `seed_start + repeat`, enabling exact reproducibility while still testing different initial phase relationships across repeats.

N=3 main evaluation. The primary model-selection evaluation uses three agents because the planned HIL system in the next stage consists of two virtual drones and one real Raspberry Pi agent. This makes the N=3 simulation directly aligned with the next experimental stage. Three visibility topologies are tested: `all_to_all`, `chain`, and `directed_chain`.

Four natural-frequency sets are used to test increasing oscillator heterogeneity: identical [2.0, 2.0, 2.0] Hz, near-identical [1.9, 2.0, 2.1] Hz, moderate heterogeneity [1.8, 2.0, 2.2] Hz, and strong heterogeneity [1.5, 2.0, 2.3] Hz. These conditions test whether each model can synchronise not only when all agents have similar natural timing, but also when one agent is substantially slower and another is faster than the nominal 2.0 Hz rhythm. Each model is evaluated for

$$4 \text{ frequency sets} \times 3 \text{ topologies} \times 20 \text{ repeats} = 240 \text{ trials.}$$

Across five models, the N=3 evaluation comprises 1200 trials. Each trial lasts 60 s and uses independent final-evaluation seeds 2000–2019.

N=5 scalability evaluation. A supplementary N=5 evaluation tests whether the conclusions from the three-agent setting generalise to larger local-neighbour networks. This test is not intended to represent an immediate five-drone hardware experiment; rather, it is a scalability proxy for decentralised local synchronisation. Five agents are evaluated under three sparse visibility graphs: `local_ring_5`, `chain_5`, and `local_degree_2_3`. These topologies restrict each agent to local neighbours rather than global visibility, so the model must propagate timing information through the graph.

Two N=5 frequency sets are tested. The near-identical set [1.8, 1.9, 2.0, 2.1, 2.2] Hz evaluates whether a model scales when natural frequencies are close but not identical. The strong-heterogeneity set [1.5, 1.7, 2.0, 2.3, 2.5] Hz evaluates whether the model can still form a common rhythm when the network contains both slow and fast oscillators. Each model is evaluated for

$$3 \text{ topologies} \times 2 \text{ frequency sets} \times 10 \text{ repeats} = 60 \text{ trials.}$$

Across five models, the N=5 evaluation comprises 300 trials using seeds 3000–3009.

Repeatability check. A second independent final evaluation was run using seeds 4000–4019 for N=3 and 5000–5009 for N=5, with all locked parameters unchanged. This checks whether the final ranking is robust to different random initial phase conditions rather than depending on a particular seed set.

Assumptions and exclusions. This evaluation assumes ideal event delivery. A flash produced by a visible neighbour is delivered according to the topology graph without missed detections, false positives, camera latency, exposure variation, or brightness-dependent detection error. Physical drone motion, collision avoidance, flight dynamics, and geometric camera field of view are also excluded. These exclusions are intentional: This stage is designed to isolate synchronisation algorithm behaviour under controlled local-interaction graphs. Perception noise, timing latency, LED brightness, and hardware loop effects are reserved for the following HIL stage.

3.2 Raw metrics

All metrics are computed from simulated flash-event logs, oscillator histories, and final-window steady-state statistics. Flash-based metrics are emphasised because the later HIL system will observe neighbour behaviour through visual flash events rather than direct access to all internal oscillator states. Unless otherwise stated, steady-state quantities are computed over the final 10 s of each 60 s trial.

Zero-lag group synchronisation. Zero-lag group synchronisation is the strict action-triggering criterion. It requires phase synchronisation, frequency lock, and one-to-one flash lock. In this report, phase synchronisation requires pairwise timing error below 0.10 s and order parameter $R > 0.85$. Frequency lock requires final effective frequency spread below 0.05 Hz. One-to-one flash lock requires flash count ratio ≤ 1.2 .

Phase-locked group synchronisation. Phase-locked group synchronisation allows a stable non-zero offset between agents. It is useful because not all synchronised systems converge to exactly in-phase flashing. This criterion requires stable pairwise offsets with low offset jitter, while still requiring frequency consistency and one-to-one flash locking.

Convergence and steady-state metrics. Time-to-sync is the first time a sustained synchronisation criterion is achieved. Steady-state metrics are computed over the final 10 s and include mean pairwise timing error, pairwise offset jitter, final frequency spread, and order parameter R .

Flash behaviour metrics. Flash count ratio is defined as

$$\text{FCR} = \frac{\max_i n_i}{\min_i n_i}, \quad (9)$$

where n_i is the final-window flash count of agent i . FCR close to 1 indicates one-to-one flash locking. Values substantially above 1 indicate extra flashes or harmonic locking.

3.3 Weighted evaluation matrix

The final model score is an engineering design-selection score. It combines measured simulation performance with explicitly stated expert-scored categories. All category scores are normalised to $[0, 1]$.

Table 2: Composition of the weighted evaluation matrix. Sub-weights sum to 1 within each category.

Category	Internal composition	Total weight
N=3 synchronisation	0.45 zero-lag success + 0.25 phase-locked success + 0.30 one-to-one flash lock	0.30
N=3 convergence/quality	0.35 time-to-sync score + 0.25 timing-error score + 0.20 offset-jitter score + 0.20 frequency-spread score	0.15
N=3 robustness	Success averaged across frequency sets/topologies, with strong-heterogeneity performance included explicitly	0.15
N=5 scalability	N=5 zero-lag success, phase-locked success, one-to-one lock, strong heterogeneity, and local-topology performance	0.15
Hardware readiness	Expert-scored from real-time simplicity, event-based compatibility, latency/noise sensitivity, and implementation maturity	0.15
Interpretability	Expert-scored from mathematical clarity, firefly relevance, and thesis narrative value	0.10

The total score is

$$S_{\text{total}} = 0.30S_{\text{N3Sync}} + 0.15S_{\text{N3Conv}} + 0.15S_{\text{N3Robust}} + 0.15S_{\text{N5Scale}} + 0.15S_{\text{HW}} + 0.10S_{\text{Interp}}. \quad (10)$$

The hardware-readiness and interpretability scores are not treated as experimental measurements; they are expert-scored design factors and are documented separately from the raw synchronisation metrics.

4 Results and Discussion

4.1 N=3 synchronisation performance

EAPF consensus achieved 100% zero-lag group synchronisation across all 240 N=3 trials for that model. Kuramoto achieved 91.7% zero-lag success. EAPF tracker achieved 76.2%. The PCO variants achieved substantially lower strict zero-lag success, mainly due to persistent extra-flash behaviour under heterogeneous initial frequencies.

4.2 N=5 scalability

The N=5 scalability test separates models that work in the three-agent setting from models that remain reliable in larger local-neighbour networks. EAPF consensus maintained 100% zero-lag success across all 60 N=5 trials for that model. EAPF tracker retained partial scalability, while Kuramoto decreased under local N=5 topologies. Both PCO variants failed to produce reliable N=5 strict synchronisation.

Because the N=5 test still assumes ideal event delivery and graph-defined visibility, this result should be interpreted as algorithmic scalability under local-neighbour interaction graphs rather than as evidence of immediate five-drone hardware readiness.

4.3 Flash count ratio and extra-flash behaviour

Flash count ratio provides a direct diagnostic for extra flashes. EAPF consensus and Kuramoto maintained FCR close to 1.0, indicating clean one-to-one flash locking. EAPF tracker and PCO adaptive PRC showed moderate flash count imbalance. PCO simple showed the largest FCR, consistent with extra flashes from faster oscillators.

The PCO results are important because the models did not simply fail randomly. They often produced repeated near-synchronous flash clusters, but faster agents emitted additional flashes between group events. This behaviour is captured by FCR and by the representative raster plots.

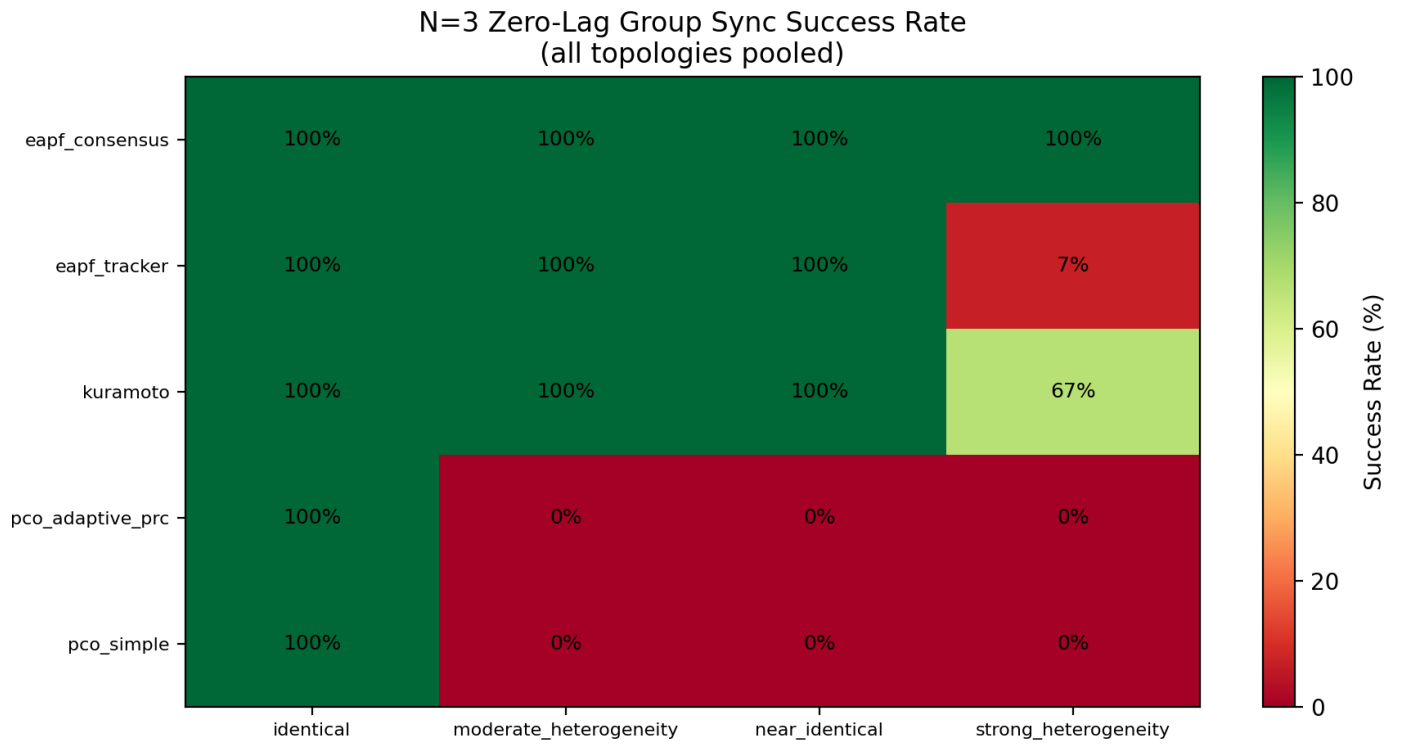


Figure 1: N=3 zero-lag group sync success rate by model and frequency set (all topologies pooled).

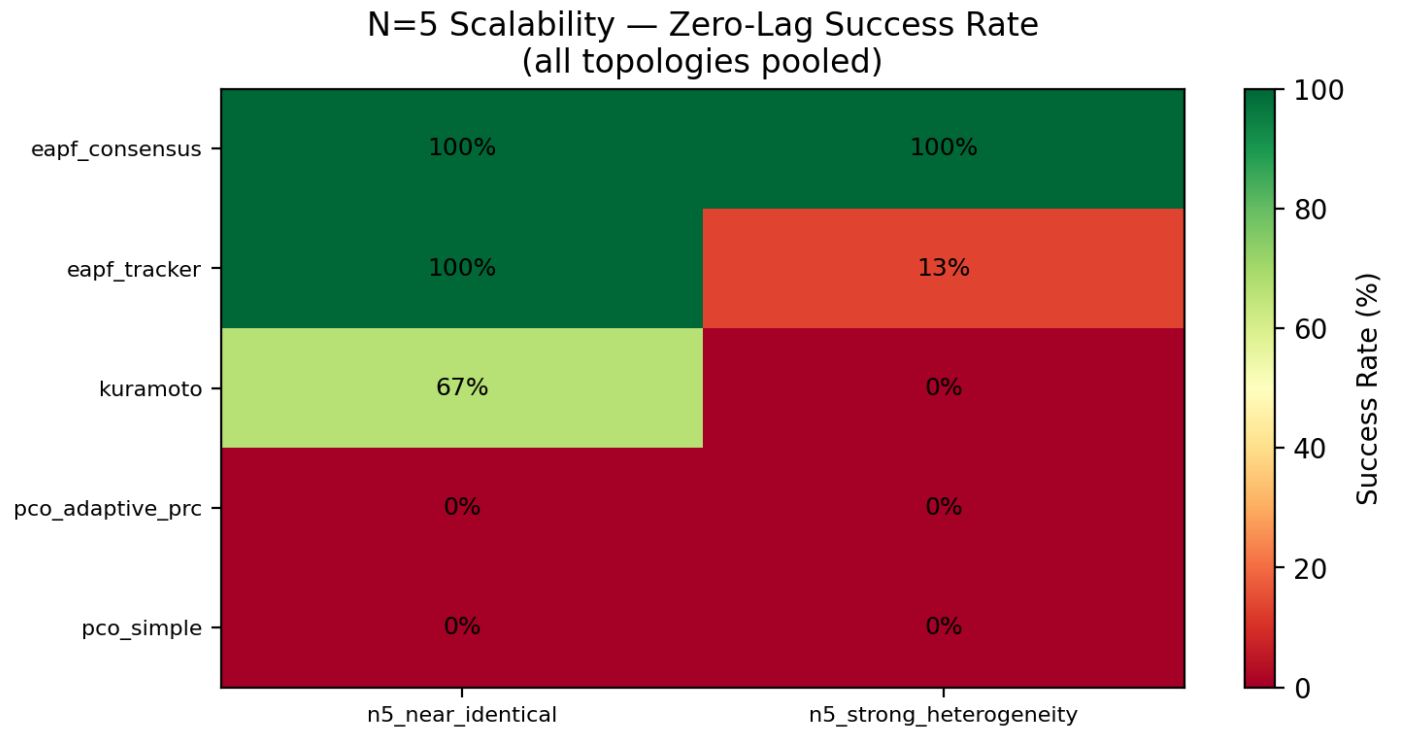


Figure 2: N=5 scalability - zero-lag success rate by model and frequency set (all topologies pooled).

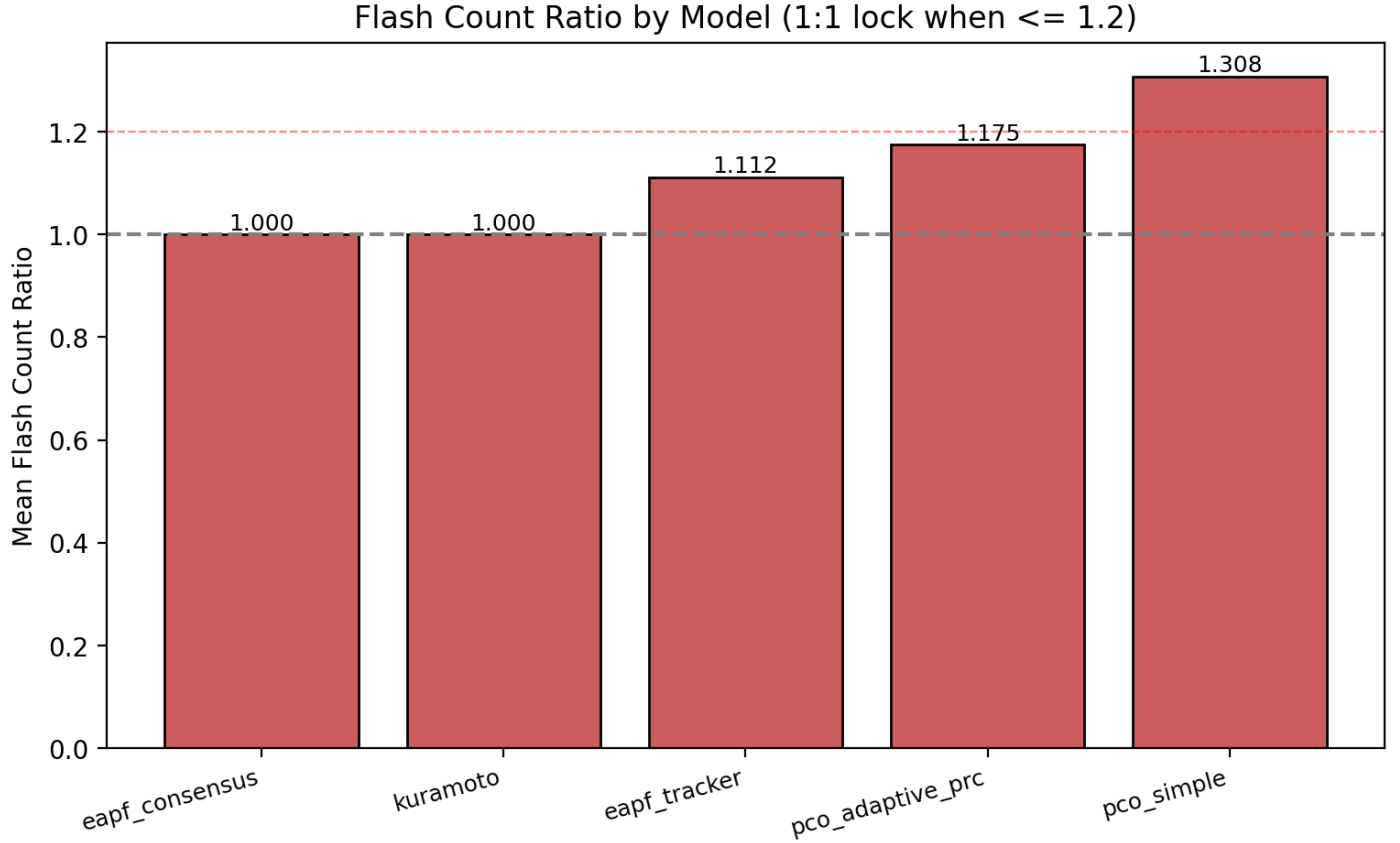


Figure 3: N=3 mean flash count ratio by model. The dashed line at 1.2 marks the 1:1 flash lock threshold.

4.4 Weighted model ranking

Table 3: Final weighted model evaluation scores. N=3 includes 1200 trials across all five models; N=5 includes 300 trials across all five models.

Model	N3 Sync	N3 Conv	N3 Robust	N5 Scale	HW	Interp.	Total
EAPF consensus	1.000	0.984	1.000	0.933	0.800	0.800	0.938
Kuramoto	0.902	0.891	0.767	0.427	0.850	0.900	0.801
EAPF tracker	0.763	0.752	0.335	0.597	0.700	0.700	0.656
PCO adaptive PRC	0.350	0.467	0.100	0.000	0.650	0.750	0.363
PCO simple	0.283	0.370	0.100	0.000	0.750	0.900	0.358

EAPF consensus ranks first with total score 0.938. It achieves the highest measured synchronisation scores and the strongest N=5 scalability. Kuramoto ranks second with total score 0.801; it is the strongest classical baseline and remains valuable as a secondary HIL comparison model. EAPF tracker ranks third and is retained as a useful event-based tracker comparison. The PCO variants are retained because they are conceptually important firefly-inspired baselines, but their low N=3 robustness and zero N=5 scalability prevent them from being selected for HIL at this stage.

4.5 Repeatability

The final ranking is robust to a change in random seeds. A second final evaluation used seeds 4000–4019 for N=3 and 5000–5009 for N=5. The ranking remained unchanged and the maximum total-score difference was 0.0039.

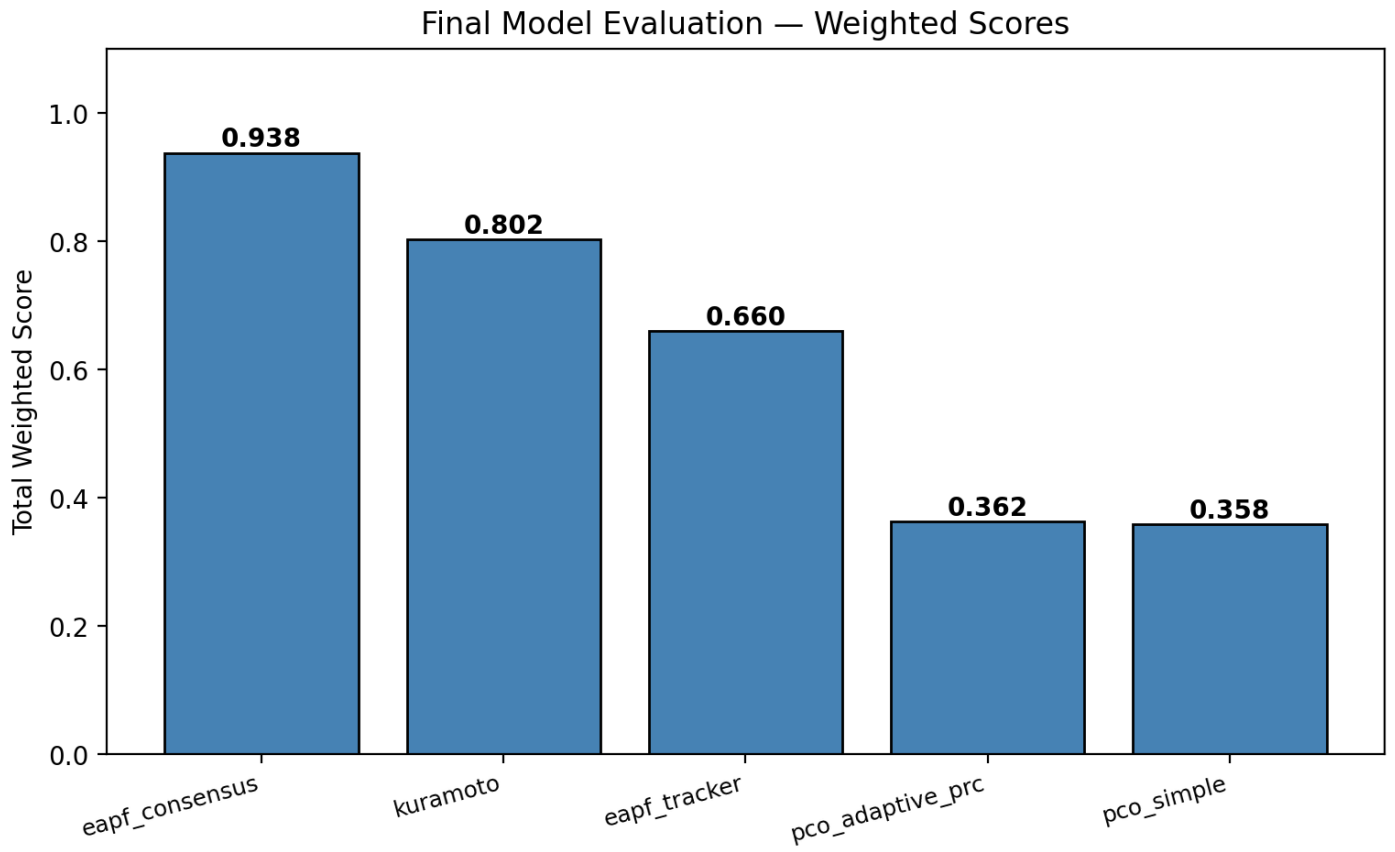


Figure 4: Total weighted model score comparison.

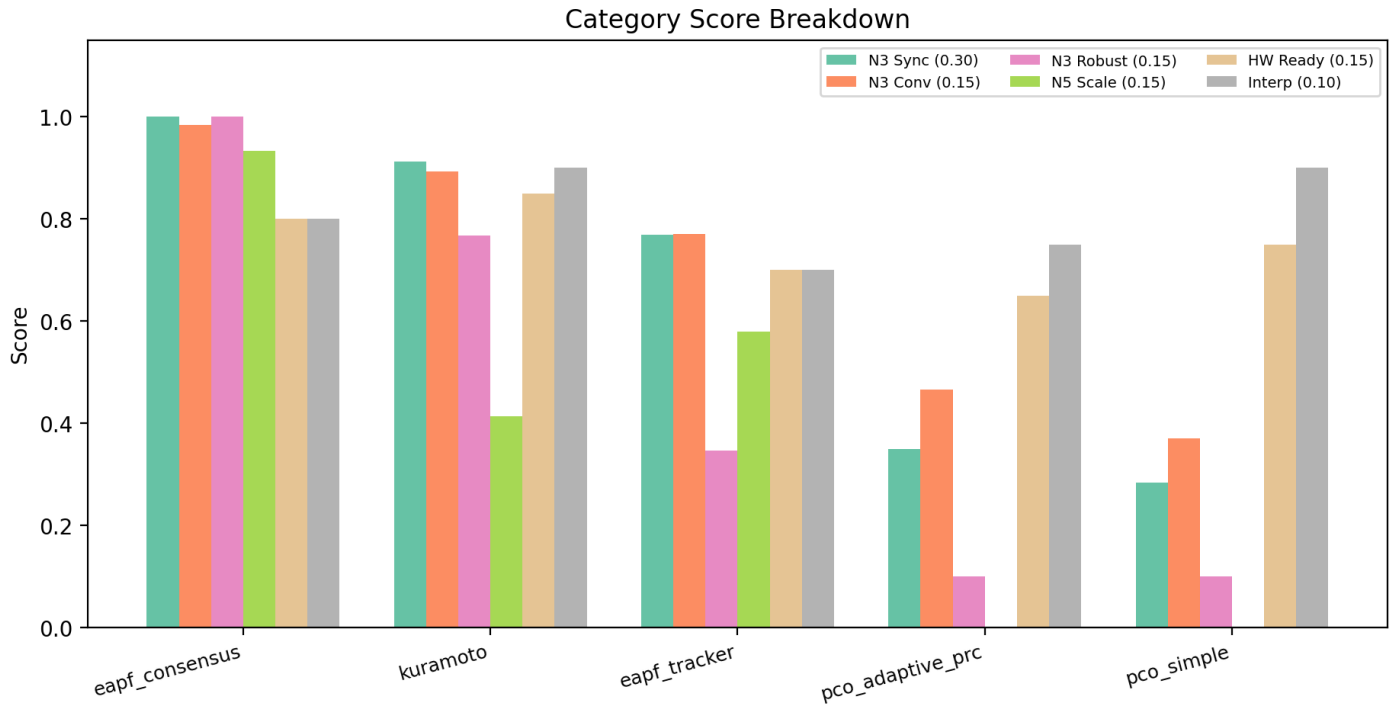


Figure 5: Category score breakdown: N3 Sync (0.30), N3 Conv (0.15), N3 Robust (0.15), N5 Scale (0.15), HW Ready (0.15), Interp (0.10).

Table 4: Repeatability across two independent final evaluations.

Model	Run 1	Run 2	Δ
EAPF consensus	0.9375	0.9375	0.0000
Kuramoto	0.8008	0.8020	+0.0012
EAPF tracker	0.6563	0.6602	+0.0039
PCO adaptive PRC	0.3625	0.3625	0.0000
PCO simple	0.3579	0.3579	0.0000

4.6 Representative raster interpretation

The representative raster plots provide a qualitative check on the aggregate metrics. EAPF consensus and Kuramoto exhibit clean one-to-one flash locking after convergence. EAPF tracker shows less robust alignment. PCO adaptive PRC and PCO simple show extra flashes, especially from faster oscillators, which explains why their phase clustering does not translate into strict group synchronisation.

Representative Flash Rasters — all_to_all, strong heterogeneity, 0-60 s

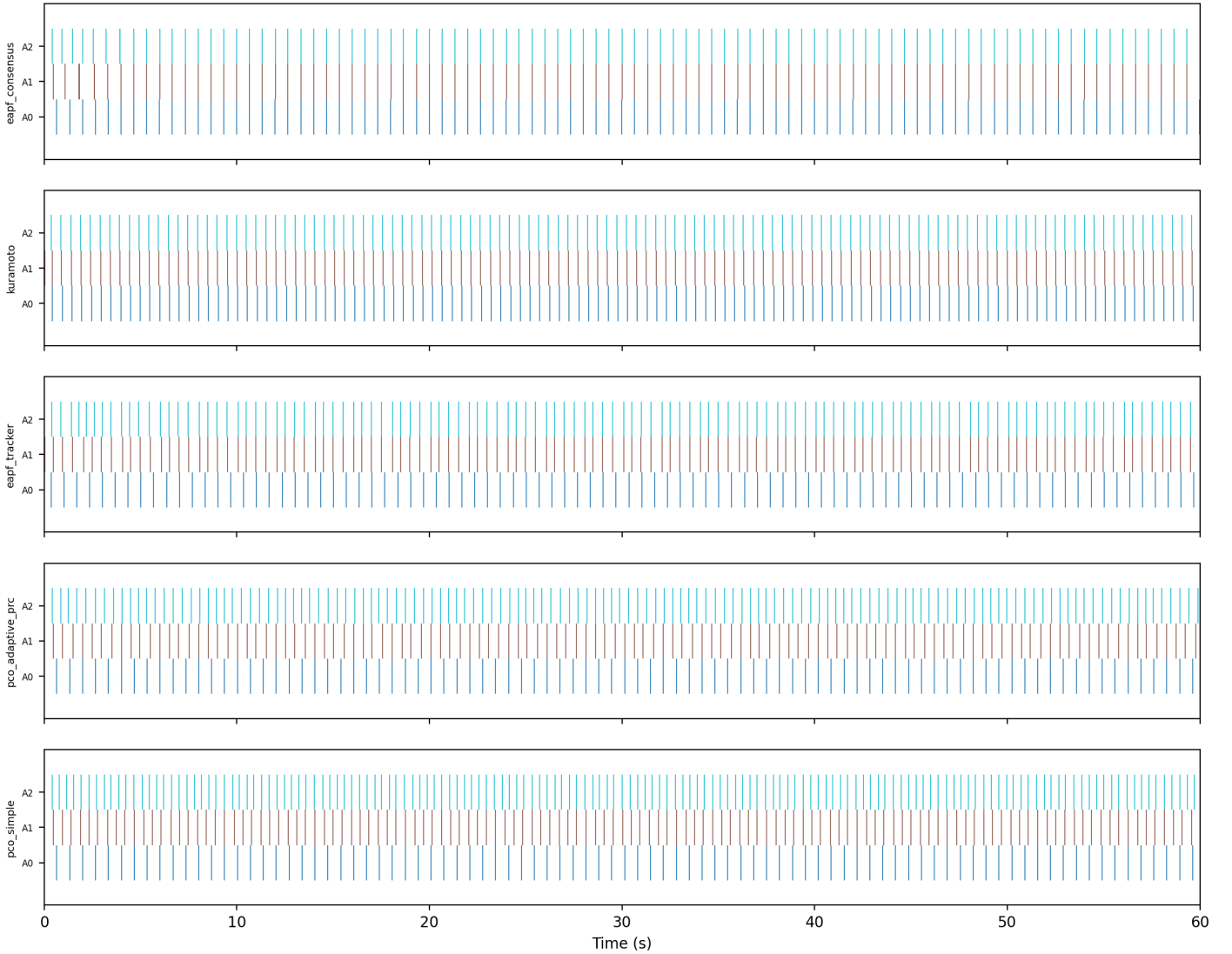


Figure 6: Representative flash rasters for all five models (all to all, strong heterogeneity, full 0–60 s). Each panel shows agent flash events as vertical tick marks. (a) Kuramoto, (b) EAPF consensus, (c) EAPF tracker, (d) PCO adaptive PRC, (e) PCO simple. Extra flashes from faster agents are visible in panels (d) and (e)..

Representative Flash Rasters — 0-10 s zoom

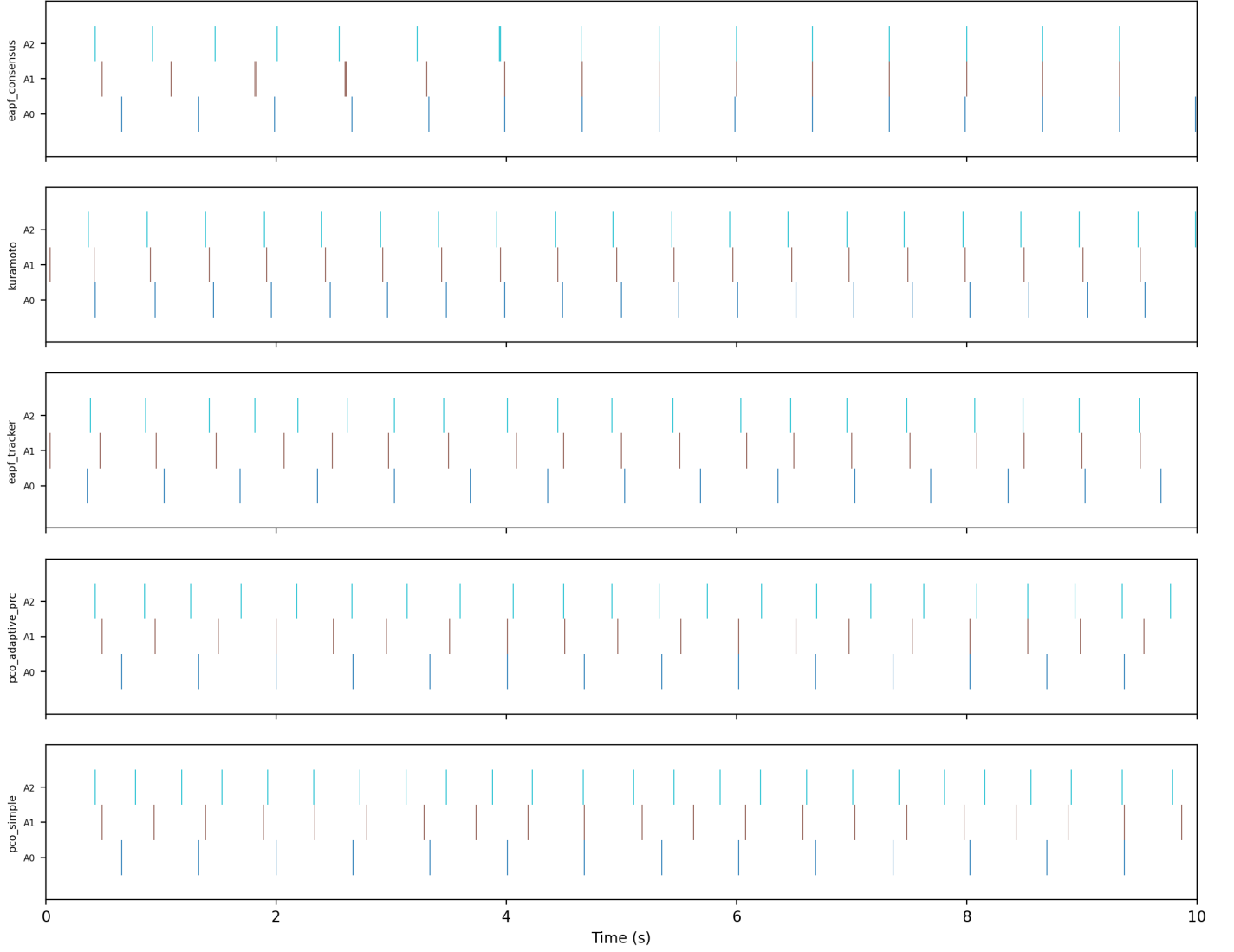


Figure 7: Representative flash rasters — 0–10 s zoom showing early convergence dynamics. Kuramoto (a) and EAPF consensus (b) converge rapidly; PCO variants (d,e) show extra flashes from the start.

Raster plots should be interpreted together with the quantitative metrics. The 0–60 s raster shows overall structure, but small early offsets can be visually compressed. The 0–10 s zoom gives a clearer view of initial convergence, while time-to-sync, FCR, frequency spread, and timing-error metrics provide quantitative confirmation.

4.7 Interpretation and limitations

Why EAPF consensus outperforms the other models. EAPF consensus corrects both phase and frequency using local neighbour estimates derived from flash timestamps. The low locked gains, $g_p = g_f = 0.02$, lead to slow but stable convergence without overshoot or correction-induced extra flashes. This explains its strong performance under both N=3 and N=5 conditions.

Why Kuramoto remains a strong baseline. Kuramoto is compact and mathematically clear, and it performs well in N=3 simulations. Its lower N=5 score is attributed to reduced robustness under local-neighbour and sparse topologies. It remains the most appropriate secondary comparison model because of its interpretability and previous single-leader visual-loop validation.

Why PCO variants remain limited. The PCO variants are important because they are the most direct firefly-inspired models. However, pulse-based phase correction alone is insufficient for reliable strict one-to-one synchronisation under strong frequency heterogeneity. Adaptive PRC reduces flash imbalance but does not fully eliminate extra flashes or restore N=5 scalability.

Current limitations. This stage is simulation-based. It does not yet include real camera latency, missed detections, exposure effects, LED brightness variation, or flight dynamics. N=5 is a scalability proxy rather than a physical five-drone experiment. Hardware readiness and interpretability scores are expert-assigned design factors rather than direct measurements. Therefore, the selected model must still be validated in Stage 4B HIL testing.

5 Conclusion

Under the tested conditions, EAPF consensus is selected as the primary HIL candidate. It achieved the highest weighted score, 100% zero-lag synchronisation across all 240 N=3 trials for that model, and 100% zero-lag synchronisation across all 60 N=5 trials for that model. Kuramoto is selected as the secondary comparison model because it is mathematically clear, performs strongly in N=3, and remains a robust classical baseline. EAPF tracker, PCO adaptive PRC, and PCO simple are retained as comparative models with identified limitations.

The next step is: implementation of a 2-virtual-drone plus 1-real-Pi HIL synchronisation experiment, using EAPF consensus as the primary model and Kuramoto as the secondary comparison baseline.

References

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